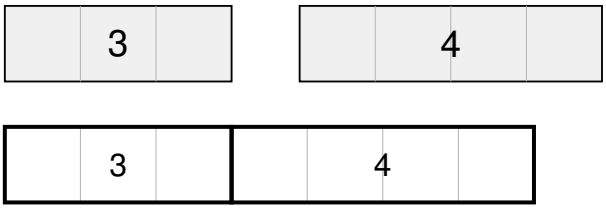
Put whole rods end to end. Leave no gaps or overlaps. You may use a length more than once. Changing the order of the same rods does not make a new way.



Problem 1: For 3-rods and 4-rods, find every impossible positive length. Explain why your list is complete.

1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	

Problem 2: Find the last impossible length for each pair: 3 and 5; 4 and 5; 4 and 7. For each pair, give a finite collection of builds that settles every longer target.

3 and 5

4 and 5

4 and 7